Gravitational Wave Signal Extraction Against Non-Stationary Instrumental Noises with Deep Neural Network

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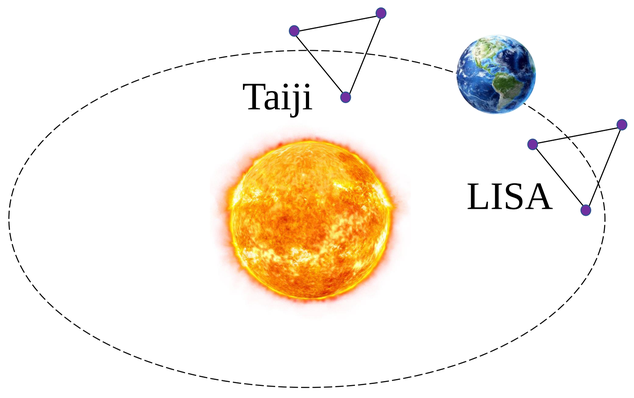
Motivations

Gravitational wave (GW) detections in space would be more seriously affected by anomalies or non-stationarities in the science measurements.

- Data gap: caused by routine maintenance or hardware problems.
- Glitch: polluted scientific data leads to false positives.
- Time-varying noise auto-correlations: caused by antenna re-orientations or payloads calibrations.

In order to obtain reliable scientific returns from observational data, it will be crucial to **extract GW signals from non-stationary data**.

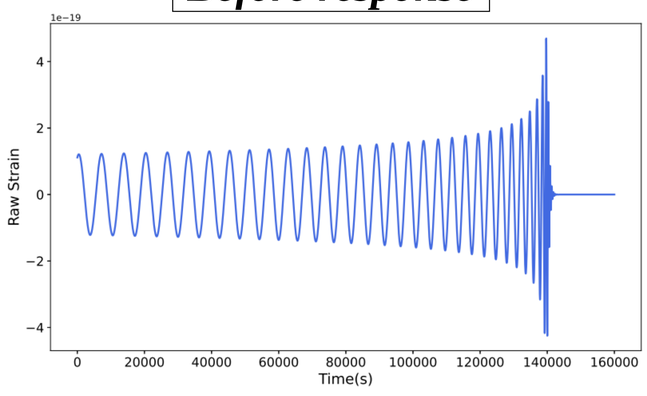


Data preparation

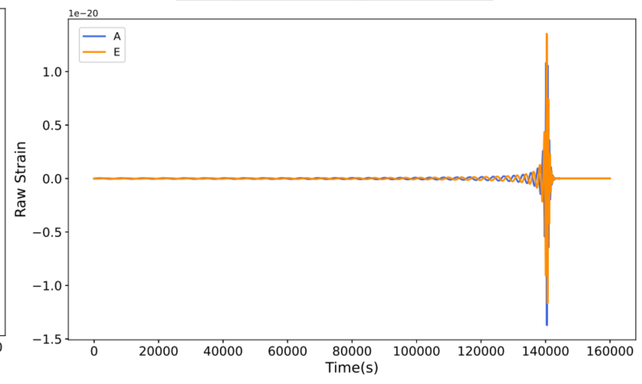
Template data generation:

The PyCBC and the waveform model SEOBNRv4 are employed to generate the coalescing MBHB GW templates, these data will **pass through the Taiji response function.**

Before response



After response

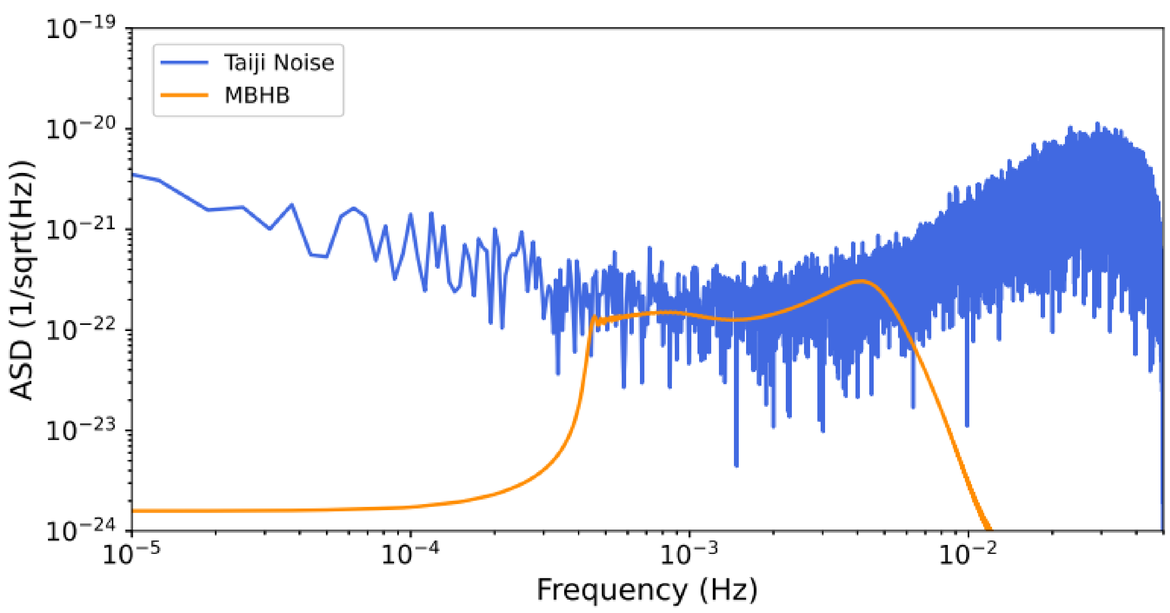


Noise generation:

The noise floor of the output data from certain Time Delay Interferometry (TDI) channel is in principle determined by **combinations of GRS residual acceleration noises and optical readout noises**.

$$S_{\mathrm{ACC}}^{1/2}(f) = 3\times10^{-15}\sqrt{1+\left(\frac{0.4\mathrm{mHz}}{f}\right)^2}\times\sqrt{1+\left(\frac{f}{8\mathrm{mHz}}\right)^4}\ \frac{\mathrm{m/s^2}}{\sqrt{\mathrm{Hz}}} \qquad S_{\mathrm{OMS}}^{1/2}(f) = 8\times10^{-12}\sqrt{1+\left(\frac{2\mathrm{mHz}}{f}\right)^4}\ \frac{\mathrm{m}}{\sqrt{\mathrm{Hz}}}$$

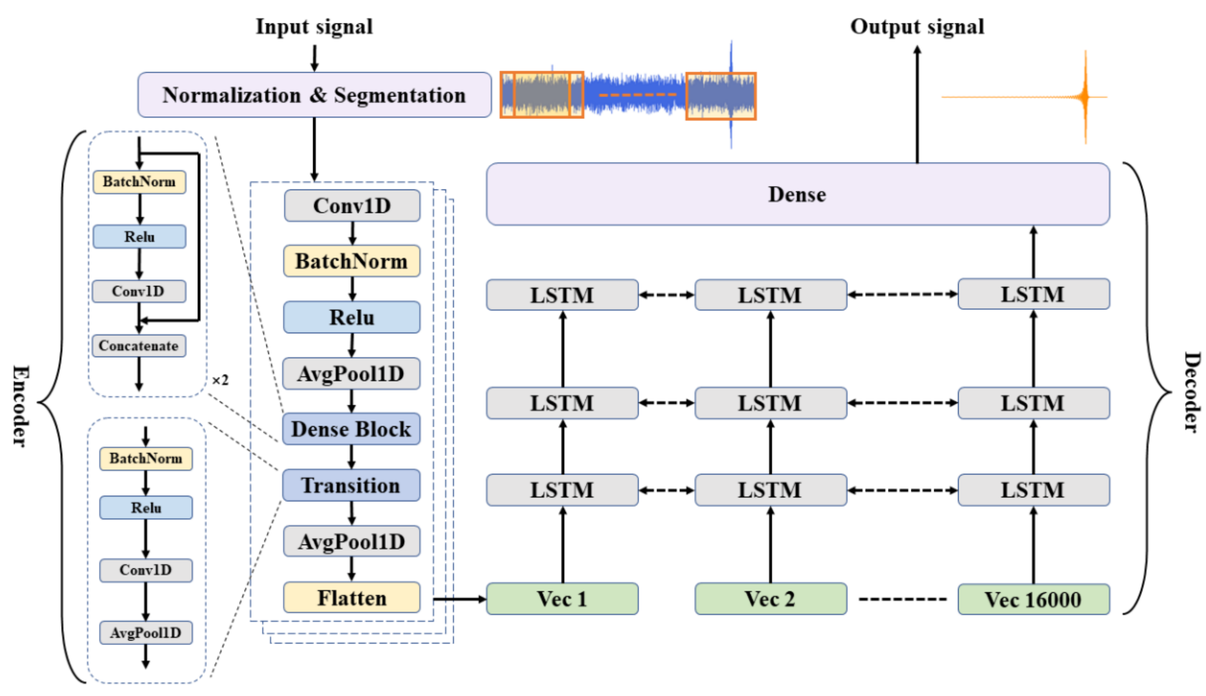
The data will look like this in the frequency domain:



Method

Denoising autoencoder composed of DenseNet and Bi-LSTM.

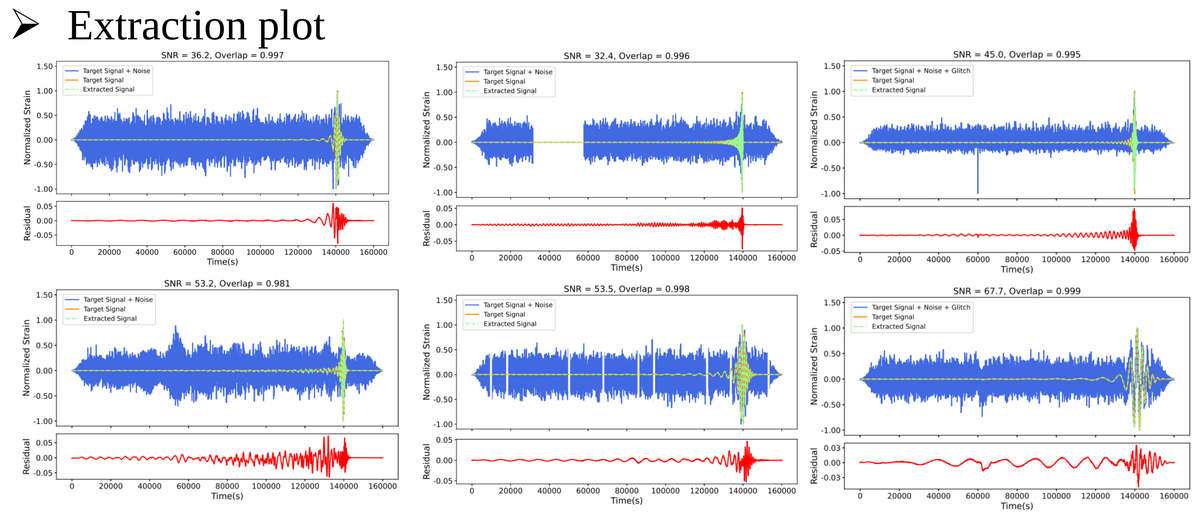
- Encoder: fully extracting the characteristics of GW sequences.
- Decoder: capturing long-term and short-term dependencies in GW sequences.



Result

Our model shows remarkable adaptability in signal extractions against all three types of non-stationarities and even their mixtures.

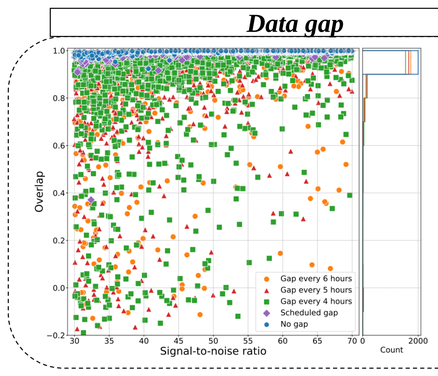
- Extraction plot



- Statistical Analysis

Evaluation Threshold: at least 80% of the extracted signals attain an overlap ≥ 0.9 with the template.

Data gap

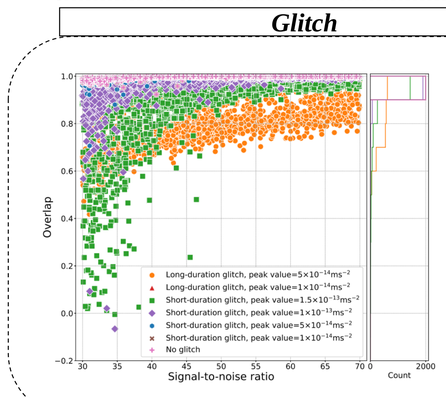


Two types of data gap

- Scheduled gap: Lasts 7 hours.
- Unscheduled gap: Lasts 6~8 mins.

Type of gap	Overlaps ≥ 0.9	Mean	Std. Dev.
No gap	100.0%	0.997	0.005
Scheduled gap	99.9%	0.996	0.016
Gap every 6 hours	86.7%	0.935	0.169
Gap every 5 hours	83.1%	0.923	0.181
Gap every 4 hours	77.3%	0.893	0.220

Glitch

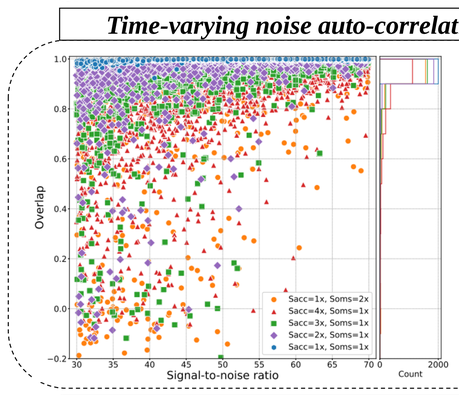


Two types of glitch

- Long-duration glitch: Lasts about 3 hours.
- Short-duration glitch : Lasts about 70 seconds.

Duration	Peak value(ms^{-2})	Overlaps ≥ 0.9	Mean	Std. Dev.
None	None	100.0%	0.997	0.005
Short	1×10^{-14}	100.0%	0.997	0.005
Short	5×10^{-14}	99.9%	0.995	0.010
Short	1×10^{-13}	94.9%	0.977	0.057
Short	1.5×10^{-13}	72.0%	0.893	0.161
Long	1×10^{-14}	100.0%	0.996	0.006
Long	5×10^{-14}	31%	0.830	0.112

Time-varying noise auto-correlations



Sacc: residual acceleration noises.
Soms: optical readout noises.

$S_{\mathrm{OMS}}(f)$	$S_{\mathrm{ACC}}(f)$	Overlaps ≥ 0.9	Mean	Std. Dev.
1x	1x	100.0%	0.997	0.005
1x	2x	92.2%	0.962	0.117
1x	3x	81.5%	0.923	0.158
1x	4x	56.0%	0.842	0.205
2x	1x	78.3%	0.894	0.224

Mixture

Data gaps		Non-stationary Gaussian noise		Glitches		Results		
Scheduled gap	Unscheduled gap	$S_{\mathrm{OMS}}(f)$	$S_{\mathrm{ACC}}(f)$	Duration	Peak value (ms^{-2})	Overlaps ≥ 0.9	Mean	Std. Dev.
✓	×	1x	3x	Short	5×10^{-14}	81.6%	0.919	0.167
✓	×	1x	3x	Long	1×10^{-14}	80.5%	0.921	0.160
×	Gap every 10 hours	1x	2x	Short	5×10^{-14}	82.1%	0.914	0.194
×	Gap every 10 hours	1x	2x	Long	1×10^{-14}	81.8%	0.905	0.216

Conclusion

- Deep learning methods have the potential to remain robust in the face of anomalies in space-based GW detections.
- Unscheduled gaps will lead to more serious spectrum leakage, and our future direction will focus on mitigating its impact.

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